Matthew Ngoy

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Summary

Computer science senior preparing for PhD research in AI security, focused on LLM red-teaming, jailbreak robustness, and verification of AI computation.

Education

Kean University Union, NJ

Bachelor of Science in Computer Science (Cybersecurity Option); GPA: 3.52 *Expected May 2027*

- Honors & Programs: NSF IUSE Scholar (2025, 2026); CRA UR2PhD; CAHSI Travel Scholar; Dean's List; Upsilon Pi Epsilon; WELL Program; Young Envoys Scholarship.

Technical Skills

Languages: Python, Java, JavaScript, SQL, PHP, PowerShell

Research & Security: LLM benchmarking, adversarial testing, citation checking, OSINT, OWASP LLM Top 10, MITRE ATLAS

Tools: Linux (Ubuntu, Kali), Docker, Git, MySQL, SQLite, Node.js, Arduino/ESP8266, Ollama, OpenAI API

Coursework Exposure: TypeScript, Wireshark, Metasploit, Hashcat, Autopsy

Experience

Cybersecurity Research Intern May 2026 – Present

Kean University Center for Cybersecurity *Union, NJ / Wenzhou and Beijing, China*

- Investigated nonprofit cyber incidents through OSINT for the Non-Profit Cyber Incident Repository, classifying reports by CIA-triad impact, CISA sector, and TAG Cyber taxonomy.
- Taught cryptography, programming, and IP addressing across 12 lessons at the Port Authority of NY/NJ STEaM Camp, reaching 60 high school students.

Peer Mentor / Code Samurai Jan. 2025 – May 2025; Sep. 2026 – Present

Kean University Department of Computer Science and Technology *Union, NJ*

- Guide students through Python, SQL, and debugging in one-on-one sessions so they can identify and explain their next steps; returned as a paid mentor after volunteering in Spring 2025.

Secretary 2026

Women in CyberSecurity (WiCyS), Kean University Chapter *Union, NJ*

- Coordinated guest-speaker scheduling and campus event planning for PHD4FREE!, developing the research and graduate-funding event from August through October 2026.
- Presented pathways into research, paid undergraduate programs, and graduate fellowships alongside Notre Dame PhD student Demetrius Hernandez, reaching 15 attendees.

International Youth Delegate Sep. 2025

International Forum "We, the Youth" (IFWY) *New York, NY / Seoul, South Korea*

- Spearheaded project design and scope for SelectSolar, a proposal to divert solar panels from landfills and deliver low-cost, energy-efficient power solutions to educational and non-profit institutions.
- Selected for a fully funded fellowship at Cornell Tech, then voted 1 of 25 North American delegates for the IFWY Final Conference in Seoul, collaborating on international environmental policy and sustainable solutions.

Projects

LLM Red-Team Harness | *Python, SQLite, Ollama* June 2026

- Built an AI-assisted pipeline for prompt-injection, jailbreak, and data-leak tests, preserving judge verdicts and transcripts in a public archive with offline tests.

Research & Presentations

HadAgent Verification Study | *Advisor: Dr. Boyang Li* Sep. 2026 – Present

- Investigated whether verifier acceptance establishes model execution; directed the research questions and interpreted local tests designed and run by AI agents against a copied HadAgent verifier.
- Identified a score-validation gap and assessed a proposed repair through local synthetic-input tests; the repaired prototype still accepted a model claim without model execution.
- Disclosed the findings to the authors, limiting conclusions to the tested local prototype rather than deployed network behavior.

Debate Coach LLM Evaluation | *CRA UR2PhD* Feb. 2026 – Apr. 2026

- Helped fine-tune and evaluate LLM debate coaches designed to challenge users' arguments, working with two teammates.
- Audited citation reliability with teammates and AI assistance; contributed participant interviews and presented the project at the CRA UR2PhD research showcase.

Adaptive Learning with LLMs | *NSF IUSE Scholars Academy* 2025

- Prepared and checked synthetic tutoring dialogues with Node.js and monitored fine-tuning jobs; presented qualitative response comparisons as first author at Great Minds in STEM.

Gen Z Housing Demographics | *NSF IUSE Scholars Academy* 2026

- Corrected a county/state aggregation mismatch as sole student author; presented the weak county-level association at Great Minds in STEM without inferring causation.